



TreeDots: Reducing food waste and carbon footprint with Google Maps Platform

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GOOGLE MAPS PLATFORM RESULTS

- ✓ Reduced the number of service calls from customers checking on driver ETAs by 20%
- ✓ Cut delivery times and enhanced customer satisfaction
- ✓ Reduced rescheduled deliveries by 10%
- ✓ Lowered delivery failure rates from 2% to less than 1%
- ✓ Contributed to the business' eco-friendliness by reducing its carbon footprint

With Google Maps Platform, TreeDots is optimizing driver route management and location accuracy to improve delivery

services, customer experience, and loyalty.

TreeDots is a wholesaler of surplus and imperfect food supplies in Southeast Asia with about 220 employees across Singapore, Indonesia, Malaysia, and the Philippines. Surplus and imperfect foods are often sent to a landfill instead of a store, which creates environmental concerns and contributes to [food waste](#)—which accounts for about [11% of total](#) waste produced in Singapore alone.

[TreeDots](#) aims to reduce that food waste by distributing excess or imperfect food items at wholesale rates to suppliers, who then sell them on to F&B businesses and other businesses. The company operates a cold storage warehouse, a fleet of delivery vehicles, and provides dashboards for suppliers and merchants to track deliveries.

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Medhi Bechina
Director of Engineering,
TreeDots

TreeDots employs about 40 drivers in Singapore who collectively make 800 to 1,500 deliveries per day to customers across the country. However, managing a range of variables that affected delivery schedules—including traffic patterns, prompt delivery requirements for perishable items, minimizing environmental impact, and dynamic customer priorities—proved challenging.

“We needed to optimize delivery routes and adapt them on the fly, and provide greater visibility of orders to our drivers to help them manage their delivery schedules,” says Medhi Bechina, Director of Engineering at TreeDots. “[Google Maps Platform](#) could meet all our requirements.”



Hampered by downtime and increases in ad hoc requests

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Director of Engineering,
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Before deploying Google Maps Platform, the business had to establish delivery zones so its drivers could service set rosters of largely repeat customers. However, a large number of ad hoc customer requests meant drivers found it hard to juggle schedules and the business was forced to mark up to 2% of deliveries as ‘failed,’ meaning the delivery could not be made or had to be rescheduled with another driver, frustrating customers and costing TreeDots business opportunities.

“We were asking our drivers to manage ad hoc requests and remember all relevant details for individual customers, while working out traffic for each destination, and this was unrealistic,” says Bechina.

The business deployed a SaaS app to manage driver schedules, but frequent downtime incidents compromised productivity and customer experience. “We couldn’t enforce its use because it was not working properly and it did not have the full feature-set necessary to meet drivers’ needs,” explains Bechina. During these downtime periods, some drivers would have to spend an hour or so at the warehouse to capture order and delivery details, rather than leaving directly to start delivering to customers.



Frictionless integration with an order management system

After establishing ease of integration and cost as its criteria for a replacement system, TreeDots evaluated other SaaS apps. The company found Google Maps Platform was the best fit for its business and could integrate seamlessly with its order management system.

Without in-house expertise in Google Maps Platform and the bandwidth within its technology function to take on and complete the project quickly, TreeDots needed external support. TreeDots asked its Google Maps Platform account manager for recommendations and was steered toward Google Maps Platform Partner [Searce](#) for assistance.

Searce worked with the business to develop mobile apps for the iOS and Android platforms that would help drivers view and manage daily pickup and delivery schedules, with that data fed back to its transport management system via backend APIs. The Searce team also developed dashboards that enabled TreeDots to track performance against various key performance indicators.

TreeDots is now using a range of Google Maps Platform APIs to deliver its service. The [Routes API](#) enables it to identify the best routes between pickup and delivery destinations, the [Geocoding API](#) pinpoints addresses, and the [Places API](#) provides access to detailed location data. Meanwhile, drivers rely on the [Directions API](#) to navigate to and from desired locations with clear instructions. The business has also employed the [Maps JavaScript API](#) to create customized maps that visualize its delivery network, gaining visibility of its operations to improve its strategic planning and decision-making.

“Overall, our drivers love the new app and I receive a lot of positive feedback from them, including questions about offline mode and other capabilities,” says Bechina of the result.



Minimizing delays with traffic prediction capabilities

With Google Maps Platform, TreeDots is updating delivery routes in real time to minimize delays that can compromise driver schedules and customer experiences. In particular, a traffic prediction capability ensures drivers can avoid traffic jams and other incidents. Meanwhile, with the Geocoding API, the business can ensure drivers are directed to the right locations for delivery, minimizing the risk of errors that can delay deliveries.

“Our drivers had been using the consumer version of Google Maps on their phones, which had worked well,” explains Bechina. “However, with Google Maps Platform, we’ve been able to add rich route and location functionality to our driver app to increase its usability. For example, we’ve added a field that enables any driver who visits a location to upload a photo that helps other drivers identify that same spot in future. This is very helpful for them.”

“Google Maps Platform has had an incredibly positive impact on our business in a very short time, thanks to Searce's delivery and consulting capabilities. We’re looking to build on this success across the region.”

Medhi Bechina

Director of Engineering,
TreeDots



Reducing failed deliveries and service support calls

The integration of Google Maps Platform into its applications has driven a notable improvement in delivery efficiency. The proportion of failed deliveries has fallen from 2% to less than 1%, and service support calls made to check on driver ETAs have fallen by 20%. The company also experiences 10% less rescheduled deliveries.

“The reduction in support calls we’ve achieved with Google Maps Platform directly translates as time our customer success team gets back to support customers with their day-to-day orders,” explains Bechina. “Our drivers now receive considerably fewer ad hoc requests, which makes their day much easier to manage. They are much less stressed because they’re not doing the overtime they otherwise needed to finish deliveries due to last-minute changes.”

By locating delivery addresses with greater accuracy and ensuring drivers take the fastest and most direct routes to their destinations, TreeDots has reduced fuel consumption and consequently lowered the cost of delivery operations. The business has gained 30 minutes of additional time for deliveries per driver per day.

“Our improved timeliness and accuracy has also improved the experience of our customers, improving loyalty and retention, and helping us win new business,” says Bechina.

With Google Maps Platform and associated processes well-established in Singapore, TreeDots now plans to extend the platform’s use to its operations in Indonesia and Malaysia. “Google Maps Platform has had an incredibly positive impact on our business in a very short time, thanks to Searce's delivery and consulting capabilities. We’re looking to build on this success across the region,” concludes Bechina.

TreeDots is a wholesaler of surplus and imperfect food supplies in Southeast Asia with about 220 employees across Singapore, Indonesia, Malaysia, and the Philippines.

Industries: Food & Beverage

Location: Singapore

Products: [Google Maps Platform](#), [Directions API](#), [Geocoding API](#),

[Maps JavaScript API](#), [Places API](#), [Routes API](#)

About Google Maps Partner- Searce

A Google Maps Partner, Searce is a niche cloud consulting business that specializes in cloud data engineering, AI, machine learning, and advanced cloud services and technologies.

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